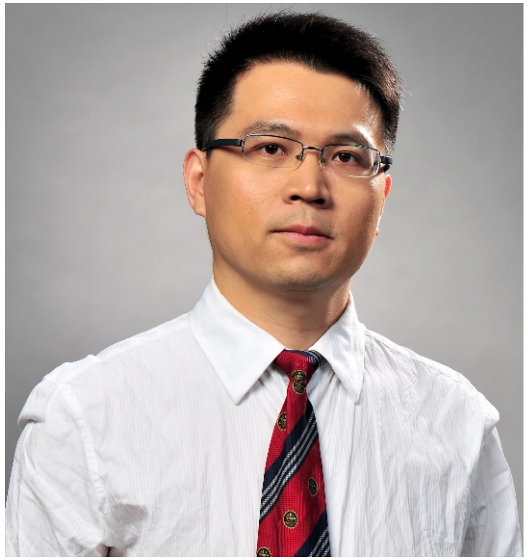
Digital Circuits
数字电路（数字电子技术）

Subject: Digital Circuits

Lecturer: 陈新

Digital Circuits



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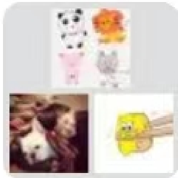
Digital Circuits



群聊：班级群 - 数电
EST2501-01

该二维码7天内(2月25日前)有效，重新进入将更新

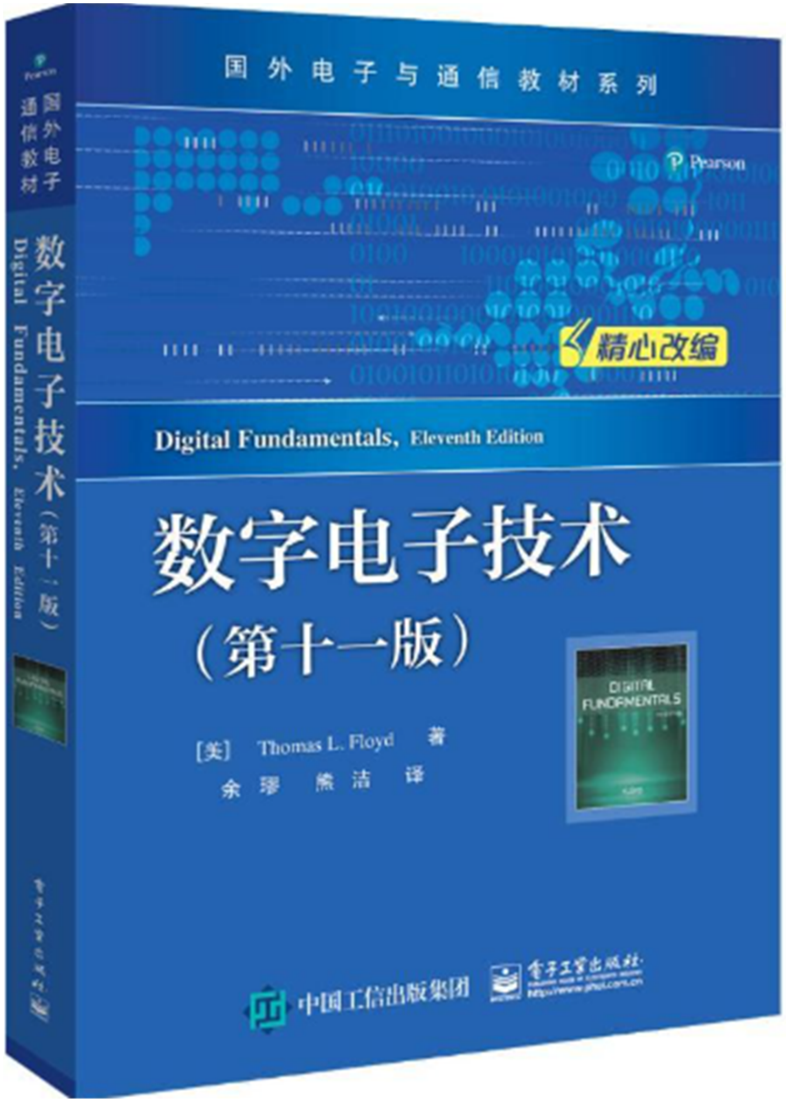
EST2501 Course Information

- Name: Digital Electronics
- Hours/Credits: 32/2.0
- Terms offered: Spring 2024
- Textbook: *Digital Fundamentals* by Floyd, 11th edition
- Grading
  - Homework Assignments, 30% (英文答题、抄题)
  - Final Exam, 70% (要求：英文试卷、不许带电子词典)

数字电子技术

- **Required:**

Digital Fundamentals 11/E by Thomas L Floyd, Science Press & Pearson Education,



CHAPTER 1

INTRODUCTORY DIGITAL CONCEPTS

1-1 DIGITAL AND ANALOG QUANTITIES

数字量和模拟量

What's Analog Quantity?

- An analog quantity(模拟量), having ==continuous== values, takes on all the infinite values in between, say, [a, b].

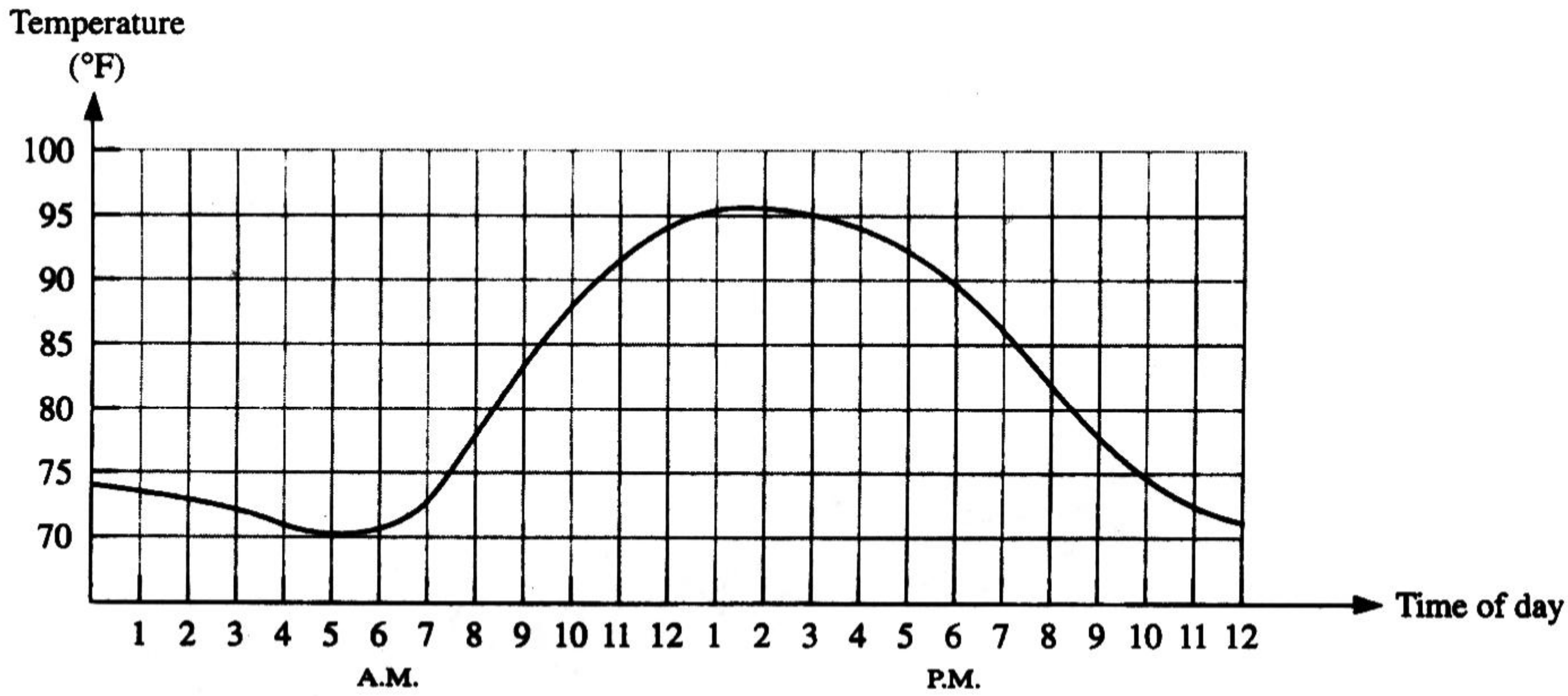


FIGURE 1–1
Graph of an analog quantity (temperature versus time).

What' s Digital Quantity?

- A digital quantity（数字量） has a set of discrete values.

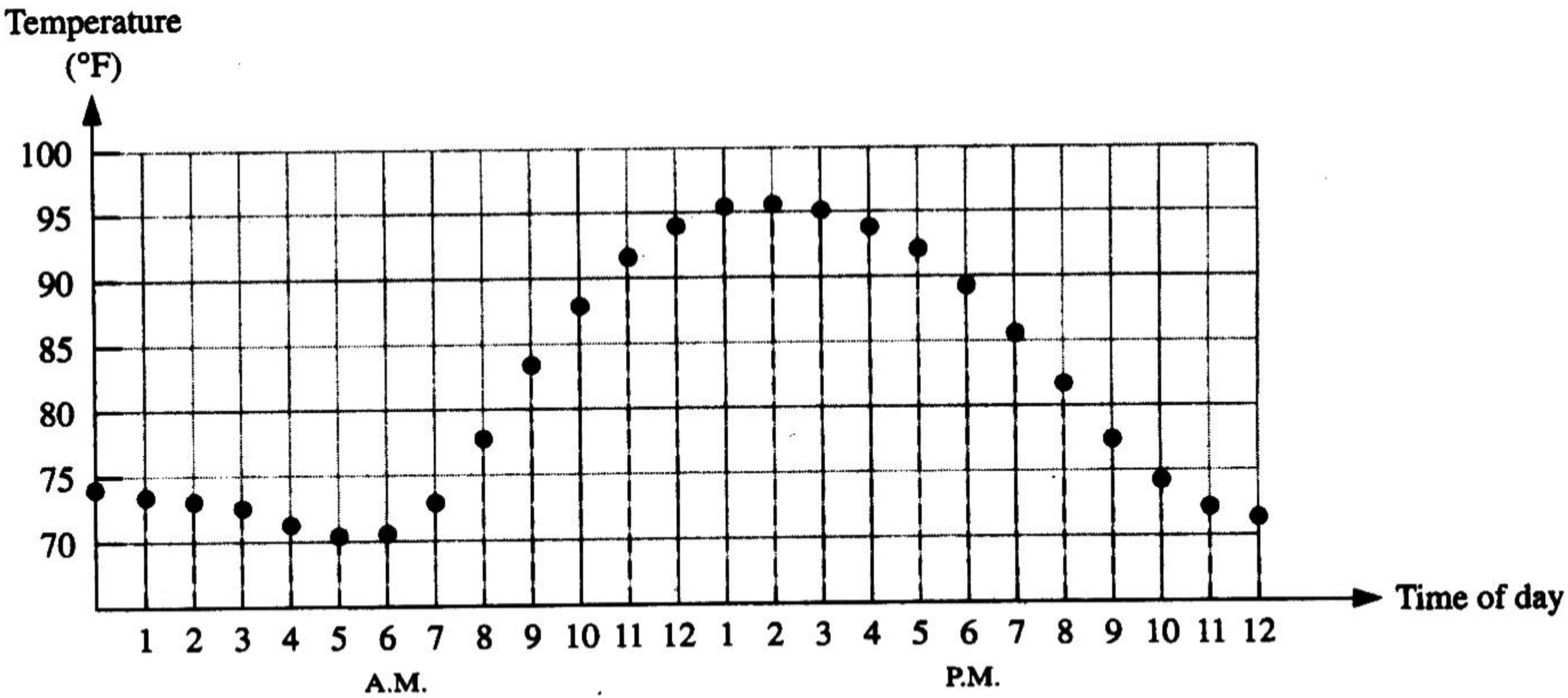


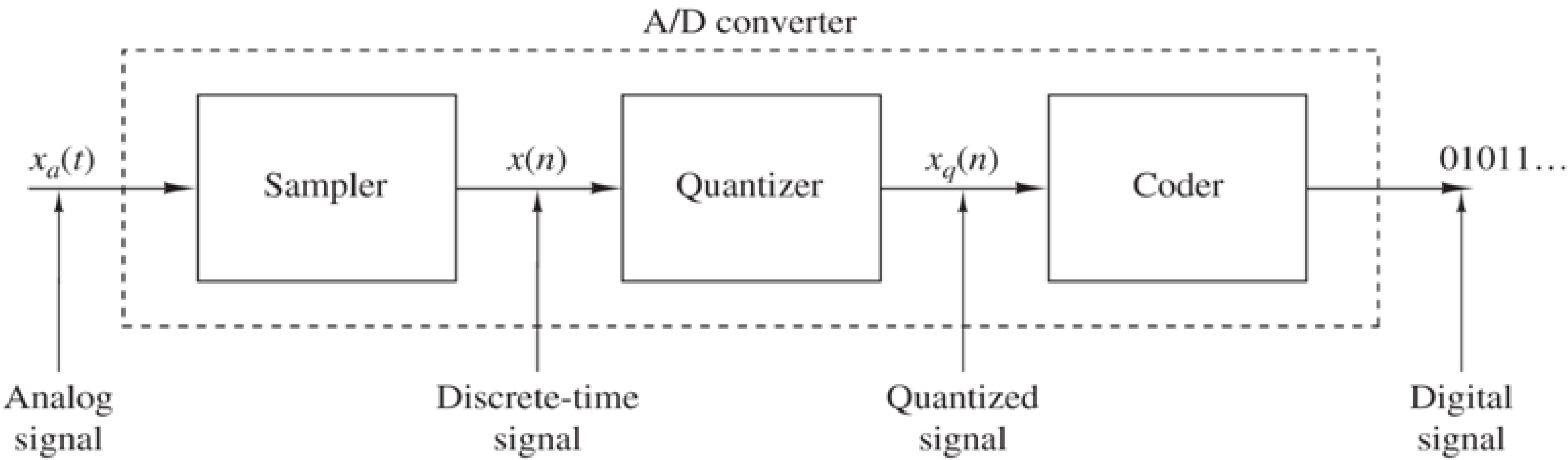
FIGURE 1–2

Sampled-value representation (quantization) of the analog quantity in Figure 1–1. Each value represented by a dot can be digitized by representing it as a digital code consisting of a series of 1s and 0s.

Analog-to-Digital Conversion

- A **signal** （信号） is a quantity that carries information.
- Mathematically, a signal is a function of some independent variables.
- To process analog signals by digital means, it is first necessary to convert them into digital form, that is, to convert them to a sequence of numbers having finite precision. This procedure is called analog-to-digital (A/D) conversion, and the corresponding devices are called A/D converters (ADCs).

- Conceptually, we view A/D conversion as a three-step process:
  - **Sampling**（采样）. The conversion of a continuous-time signal into a discrete-time signal.
  - **Quantization**（量化）. The conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal.
  - **Coding**（编码）. In this process, each quantized sample is represented by a finite number of binary digits (bits).



Basic parts of an analog-to-digital (A/D) converter.

Analog Example

- A public address system, used to amplify sound so that it can be heard by large audience, is one example of an application of analog electronics.

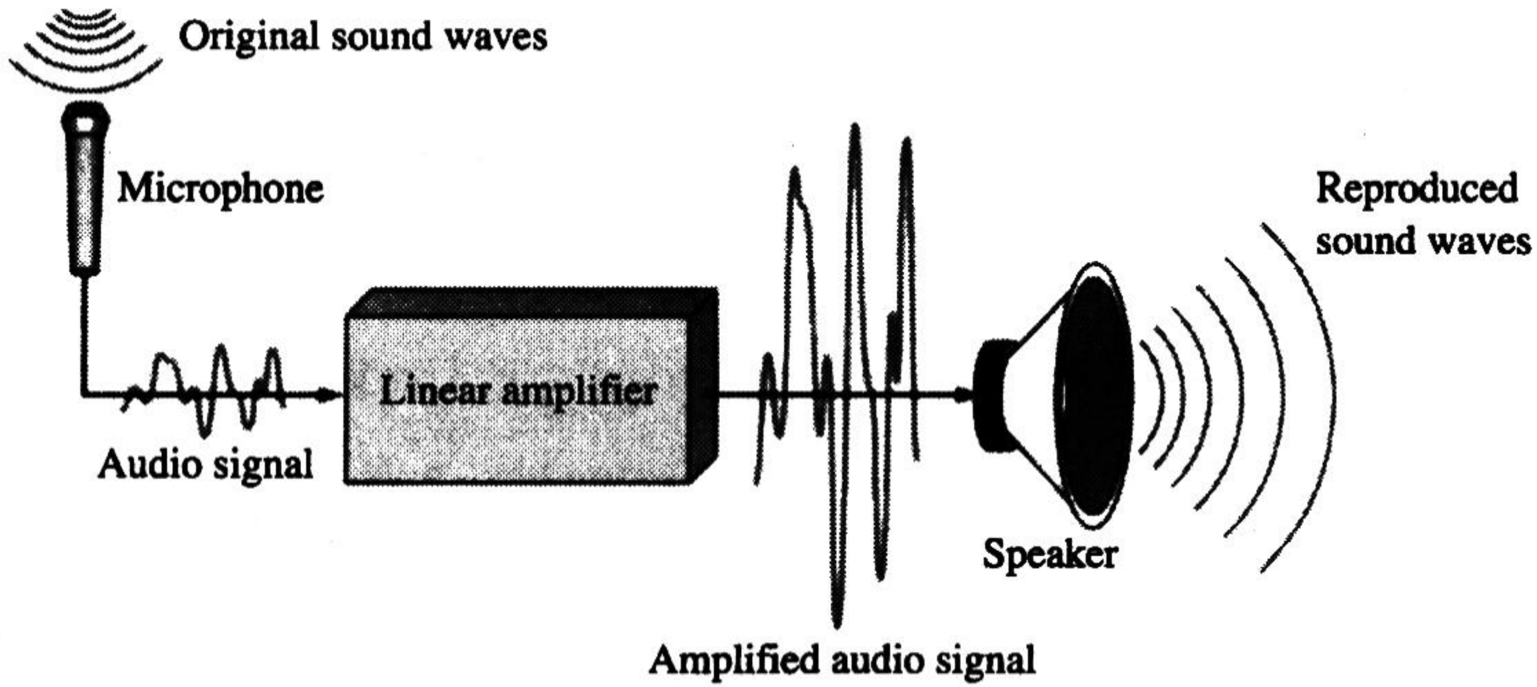


FIGURE 1–3
A basic audio public address system.

A Mixed System

- The compact disk (CD) player is an example of a system in which both digital and analog circuits are used.

Analog-to-Digital Converter (ADC)
&
Digital-to-Analog Converter (DAC)

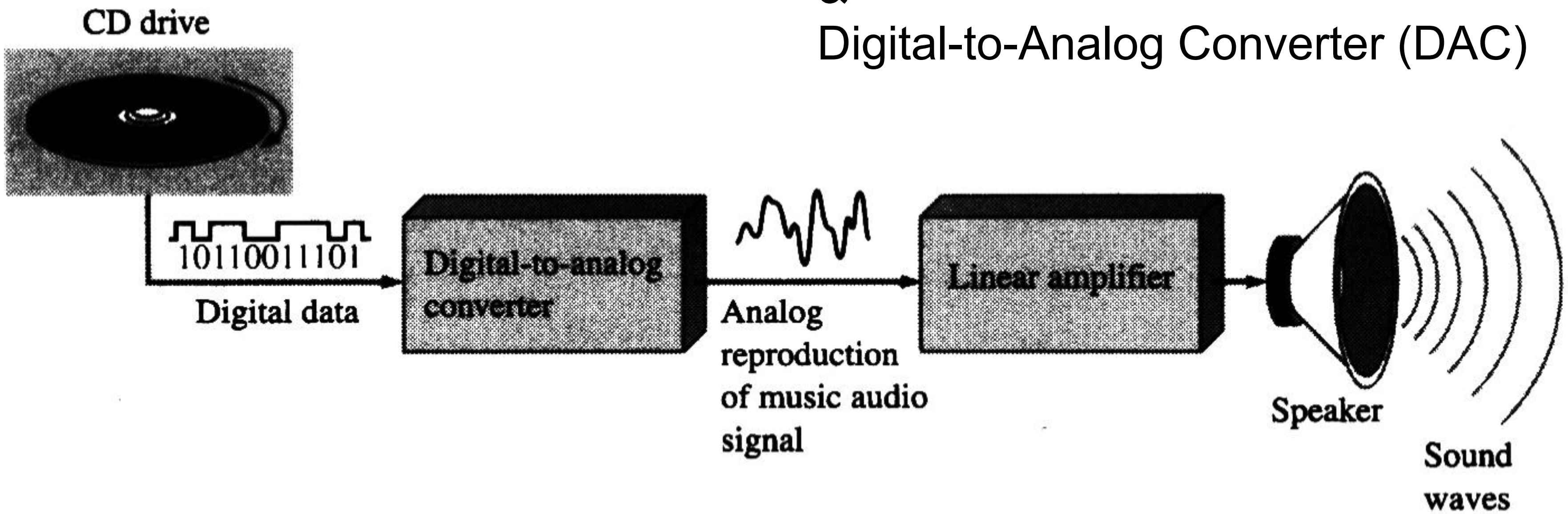


FIGURE 1–4
Basic principle of a CD player. Only one channel is shown.

Digital Example

- *A* computer system is one example of an application of digital electronics.

The Digital Advantage

- 数字电路与模拟电路比较的优势？
  - Digital has certain advantages over analog, e.g., digital data can be stored, processed, and transmitted more efficiently and reliably than analog data.
  - Digital circuits are less vulnerable to noise than analog circuits.
  - Digital circuits are more compact, less power consuming.

- Analog systems can generally handle higher power than digital systems （另外：远距离传输依然依赖模拟信号）

1-2 BINARY DIGITS, LOGIC LEVELS, and DIGITAL WAVEFORMS

二进制数、逻辑电平和数字波形

Binary Digits（二进制数字）

- The two digits in the binary system, 1 and 0, are called bits, which is a contraction of the words

0 1

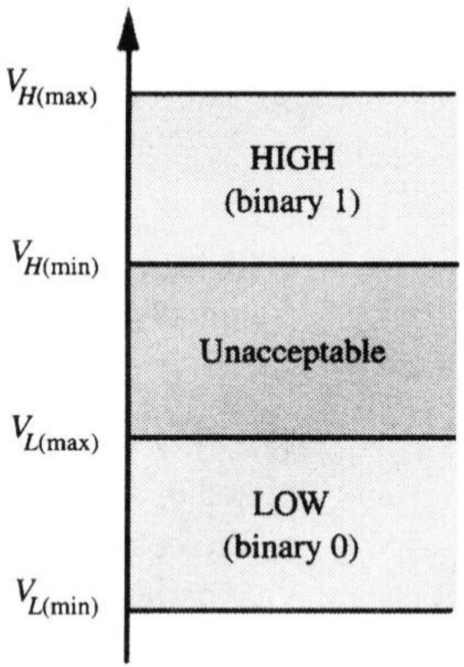
How to represent 0 and 1?

- In digital circuits, two different voltage levels are used to represent the two bits. The higher/lower voltage level is referred to as a HIGH/LOW, or H/L.

Logic Levels（逻辑电平）

- In a practical digital circuit, a HIGH can be any voltage between a specified minimum value and a specified maximum value. Likewise, a LOW can be any voltage between a specified minimum and a specified maximum.

FIGURE 1–5
Logic level ranges of voltage for a digital circuit.



Positive Logic System（正逻辑）

- A 1 is represented by HIGH and a 0 is represented by LOW.

Negative Logic System（负逻辑）

- A 0 is represented by HIGH and a 1 is represented by LOW.

Ideal Pulses（理想脉冲）

- Digital waveform consists of voltage levels that are changing back and forth between the HIGH and LOW levels or states.

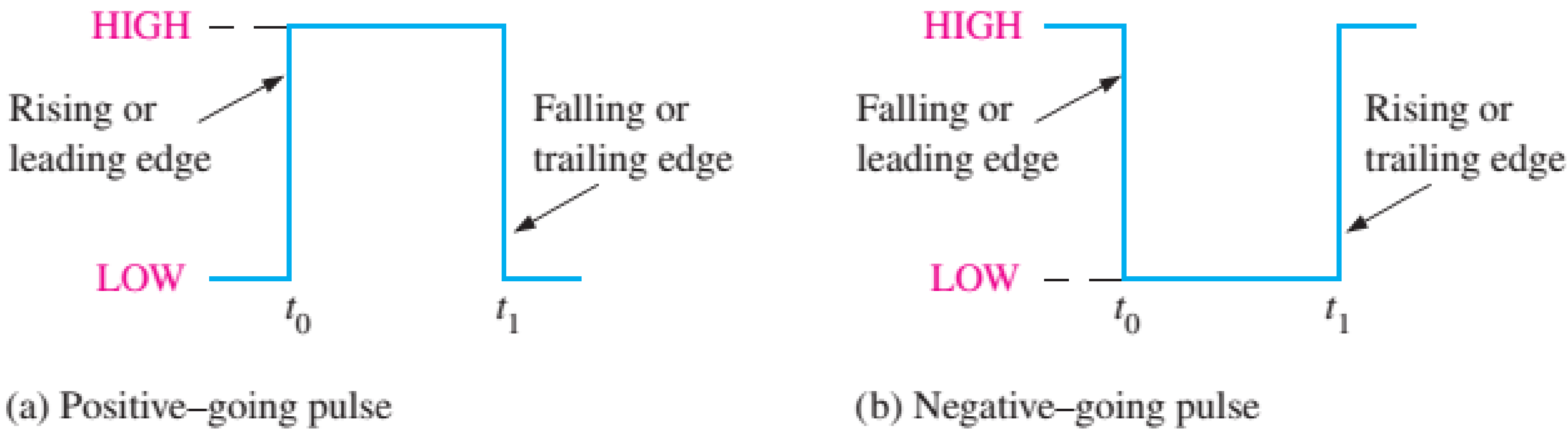


FIGURE 1–7 Ideal pulses.

Nonideal Pulse （非理想脉冲）

- The time required for the pulse to go from its LOW (HIGH) level to its HIGH (LOW) level is called the rise (fall) time. In practice, it is common to measure rise (fall) time from 10%(90%) of the pulse amplitude to 90%(10%) of the pulse amplitude.

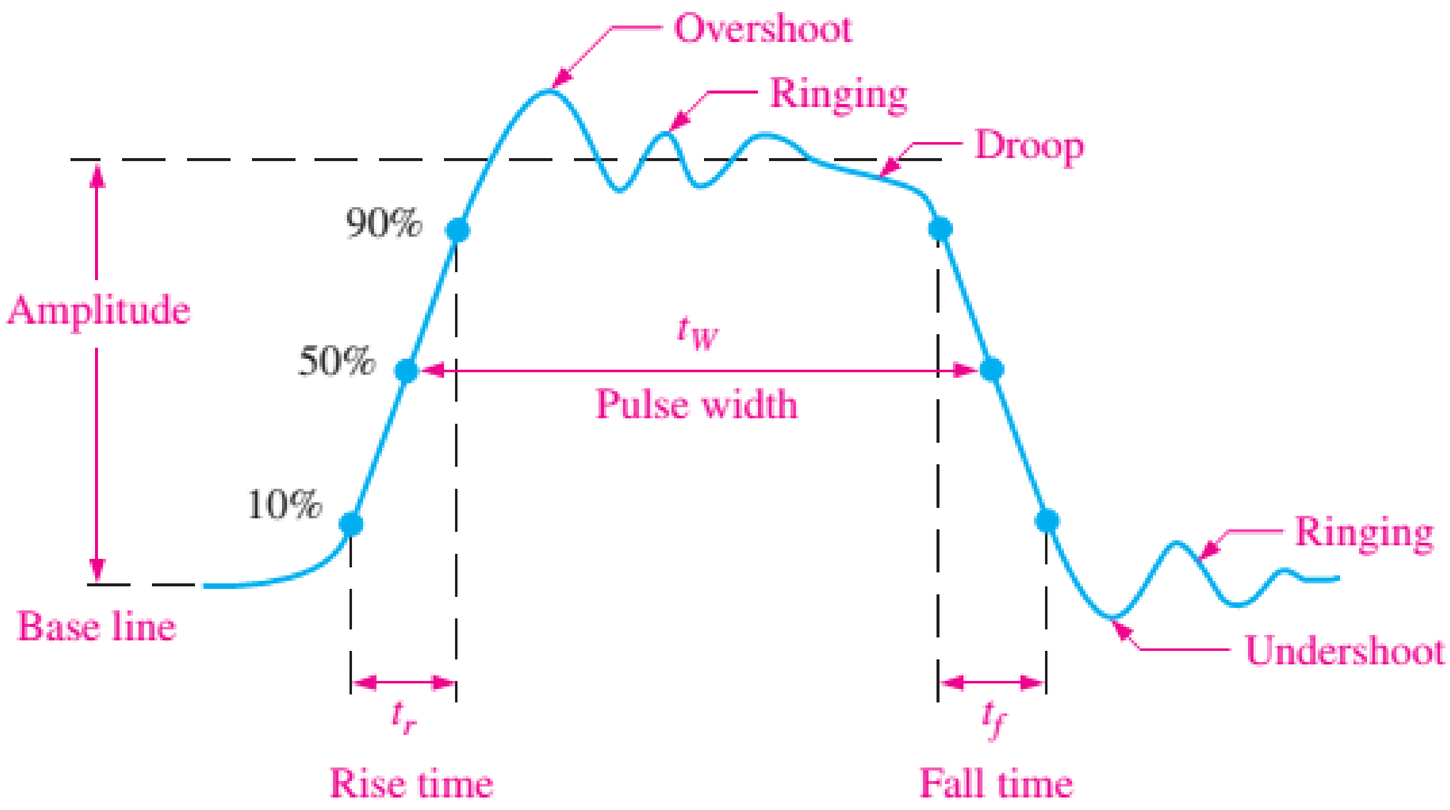
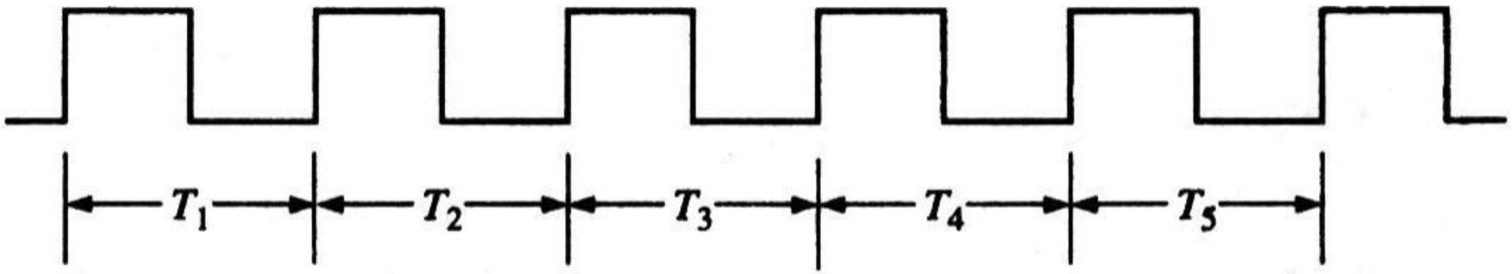


FIGURE 1–8 Nonideal pulse characteristics.

Period Pulse （周期脉冲）

- *A* periodic waveform is one that repeats itself at a fixed interval, called a period (*T*). The frequency (*f*) is the rate at which it repeats itself and is measured in hertz (Hz). The relationship between *f* and *T* is expressed as follows:

$$f = \frac{1}{T}, \quad T = \frac{1}{f}$$



Period $= T_1 = T_2 = T_3 = T_4 = T_5 = \ldots = T_n$ Frequency $= \frac{1}{T}$

(a) Periodic (square wave)

Period Pulse

- The duty cycle (D) is defined as the ratio of the pulse width (t_w) to the period (T) and can be expressed as a percentage.

$$D = \left(\frac{t_W}{T}\right)100\%$$

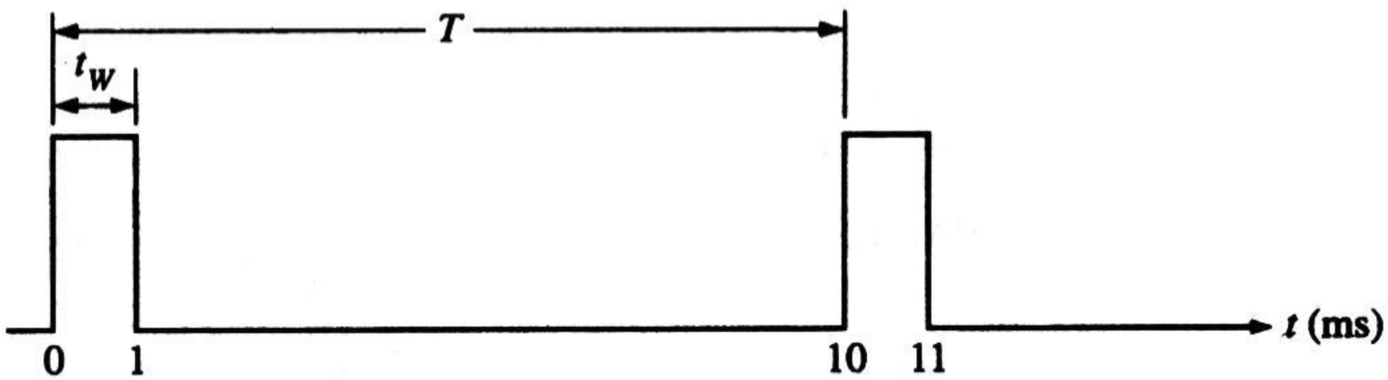
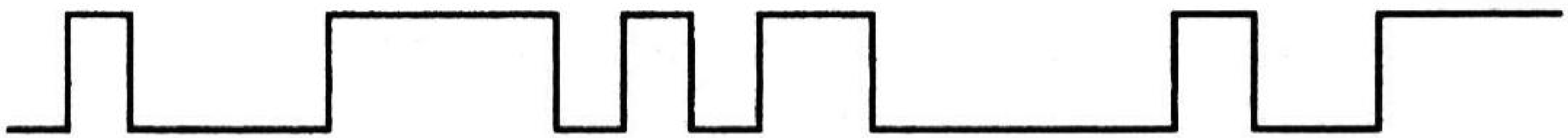


FIGURE 1–9

Nonperiod Pulse （非周期脉冲）

- A nonperiodic waveform, of course, does not repeat itself at fixed intervals and may be composed of pulses of randomly differing pulses widths and/or randomly differing time intervals between the pulses.



(b) Nonperiodic

Digital Waveforms（数字波形）

- Digital waveforms consists of a series of pulses, (voltage levels that are changing back and forth between the HIGH and LOW levels).

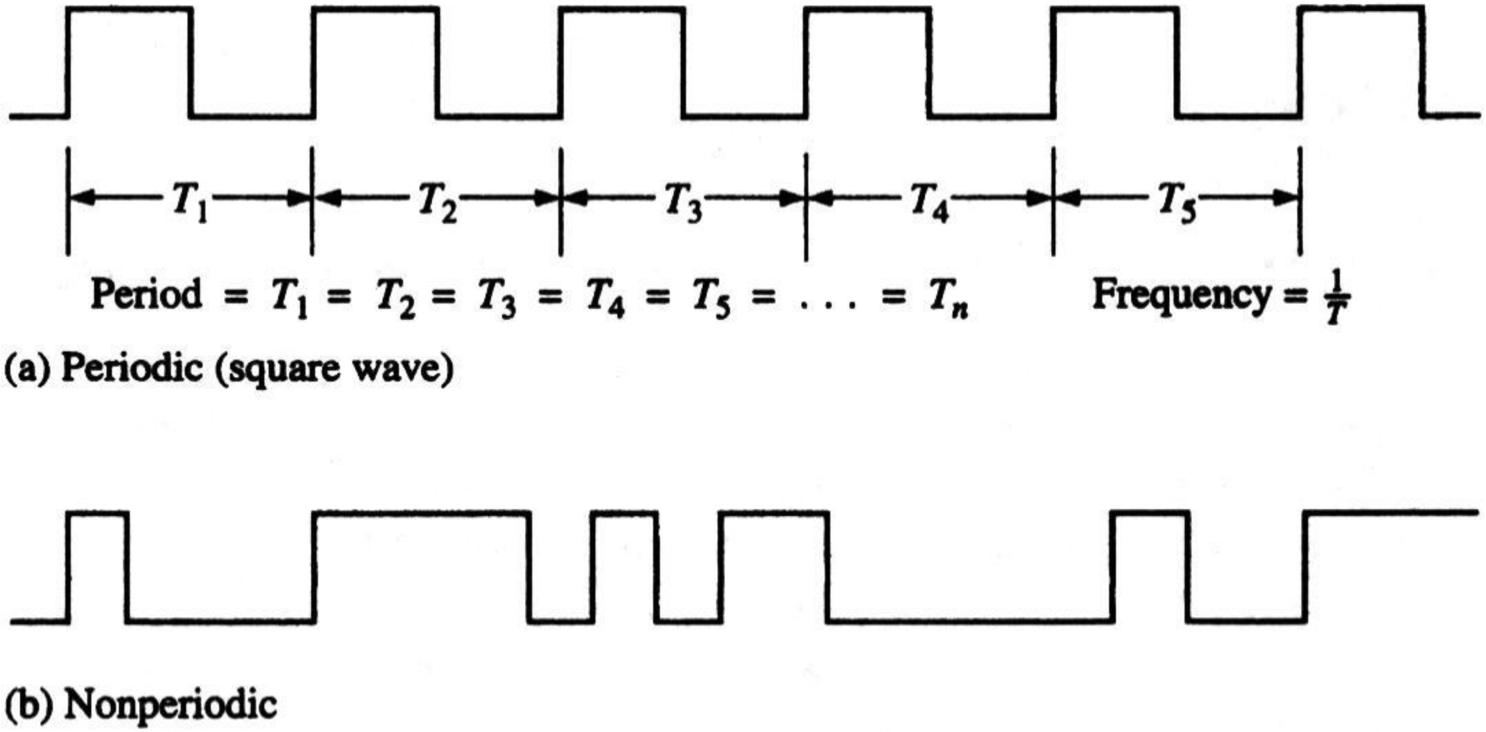
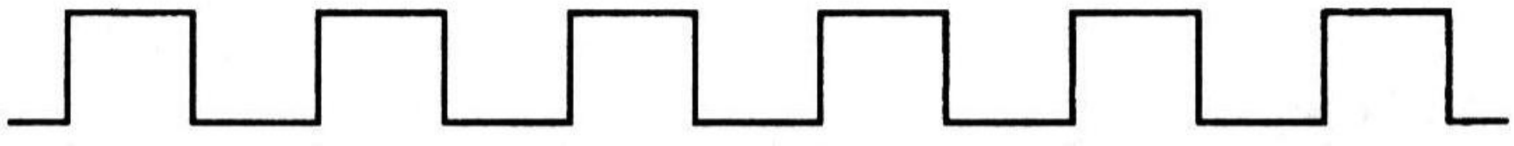


FIGURE 1–8
Examples of digital waveforms.

Pulse Train

- Digital waveforms are sometimes called pulse trains.



A Digital Waveform Carries Binary Information
（带二进制信息的数字波形）

- Binary information that is handled by digital systems appears as waveforms that represent sequences of bits. When the waveform is HIGH, a binary 1 is present; when the waveform is LOW, a binary 0 is present. Each bit in a sequence occupies a defined time interval called a bit time, or bit interval.

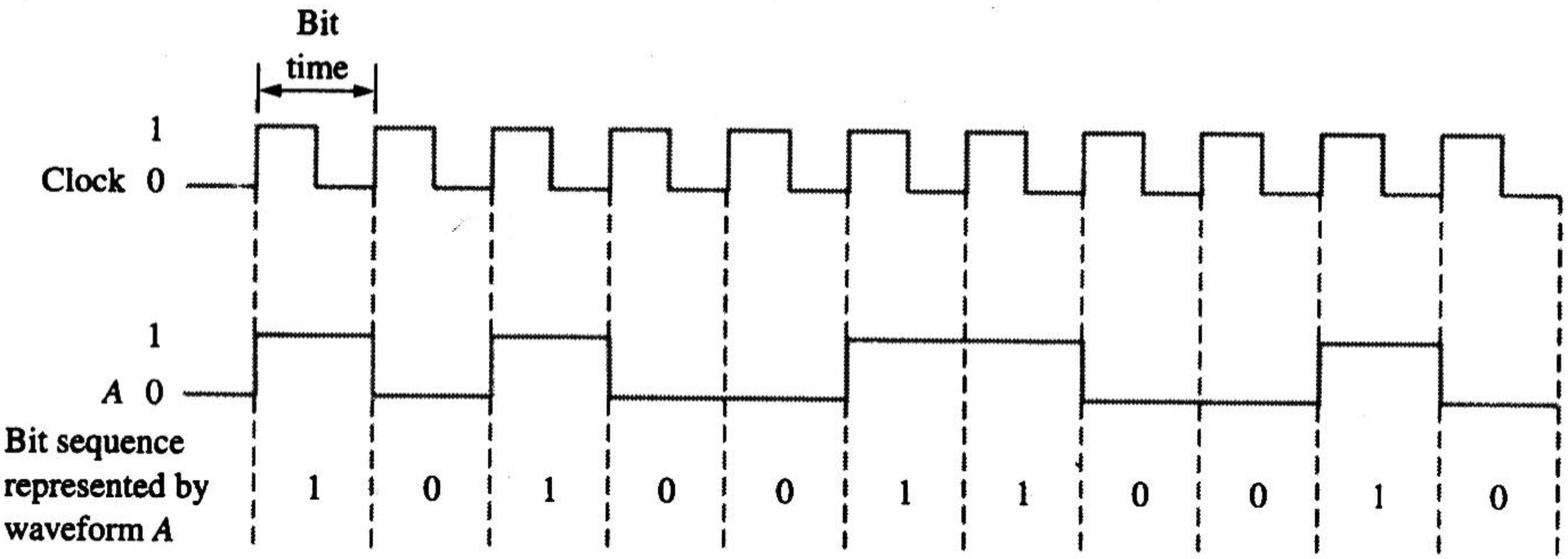


FIGURE 1–10
Example of a clock waveform synchronized with a waveform representation of a sequence of bits.

The Clock（时钟）

- In digital systems, all digital waveforms are synchronized with a basic timing waveform, called the clock. The clock is a periodic waveform. The clock waveform itself does not carry information.

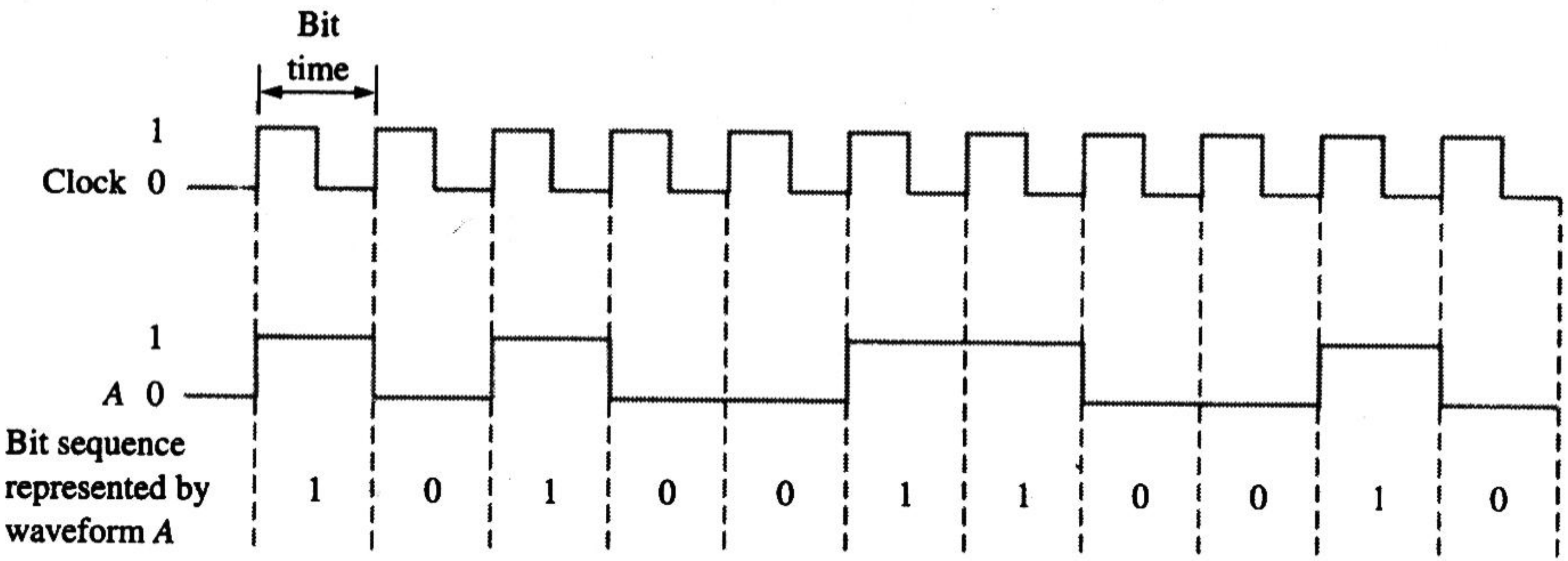
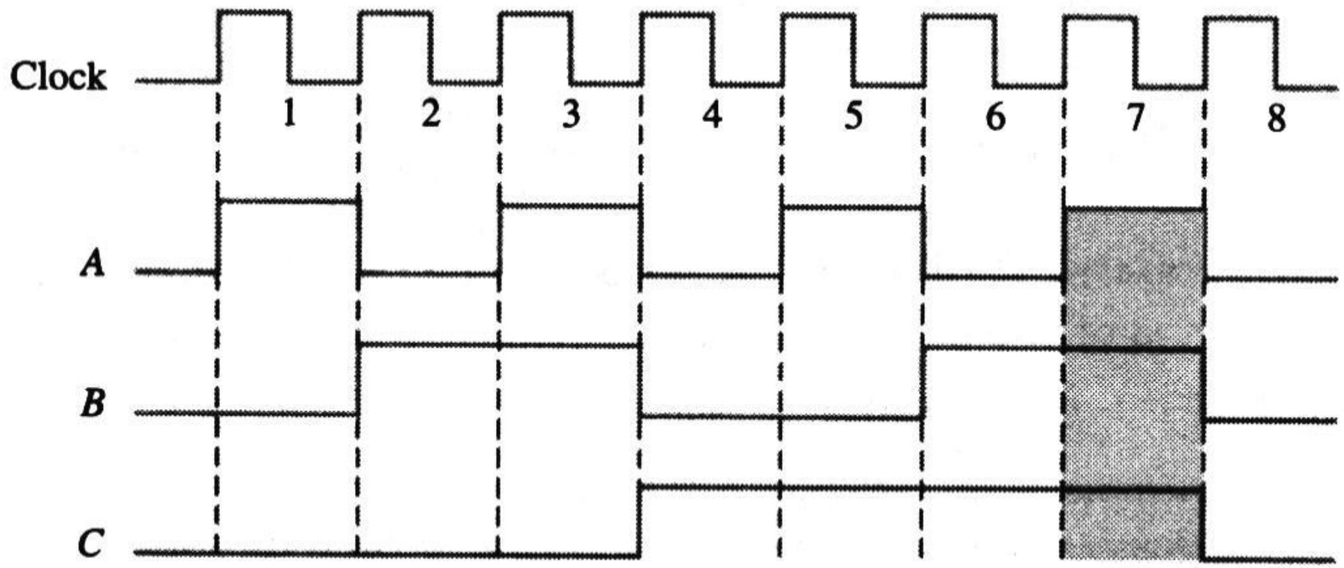


FIGURE 1–10
Example of a clock waveform synchronized with a waveform representation of a sequence of bits.

Timing Diagrams（时序图）

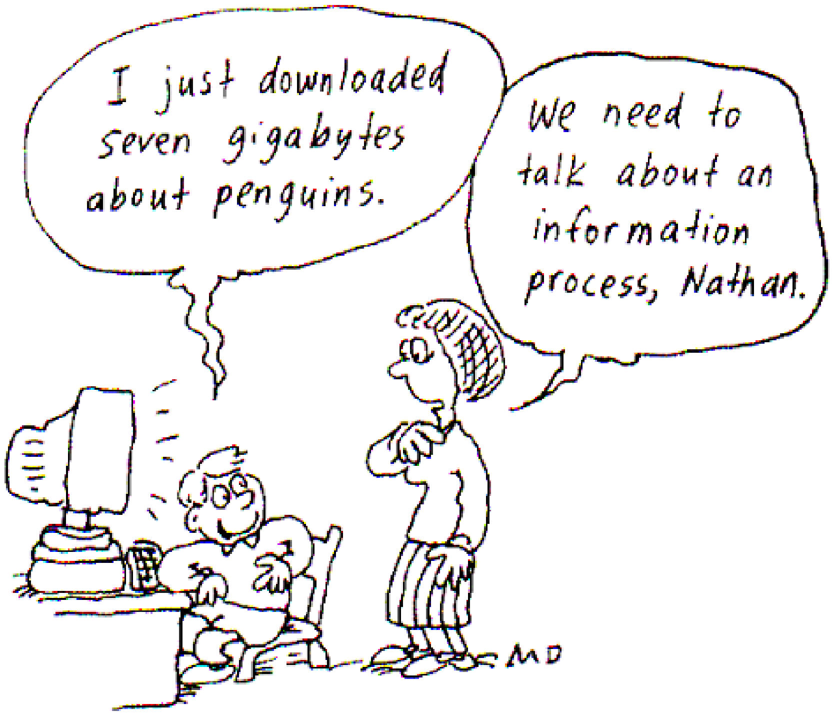
- *A timing diagram is a graph of digital waveforms showing the actual time relationship of two or more waveforms and how each changes in relation to the others.*

FIGURE 1–11
Example of a timing diagram.



Data Transfer（数据传输）

- Data refers to groups of bits that convey some type of information. Binary data, which are represented by digital waveforms, must be transferred from one circuit to another within a digital system or from one system to another in order to accomplish a given purpose.数据是指一组可以用来传达某种信息的位。



Codes（编码）

- Groups of bits (combination of 1s and 0s), called codes, are used to represent numbers, letters, symbols, instructions, and anything else required in a given application.
- The **American Standard Code for Information Interchange (ASCII)** – pronounced "askee" – is a universally accepted alphanumeric code used in most computers and other electronic equipment.

0 1 1 0 0 0 1

1

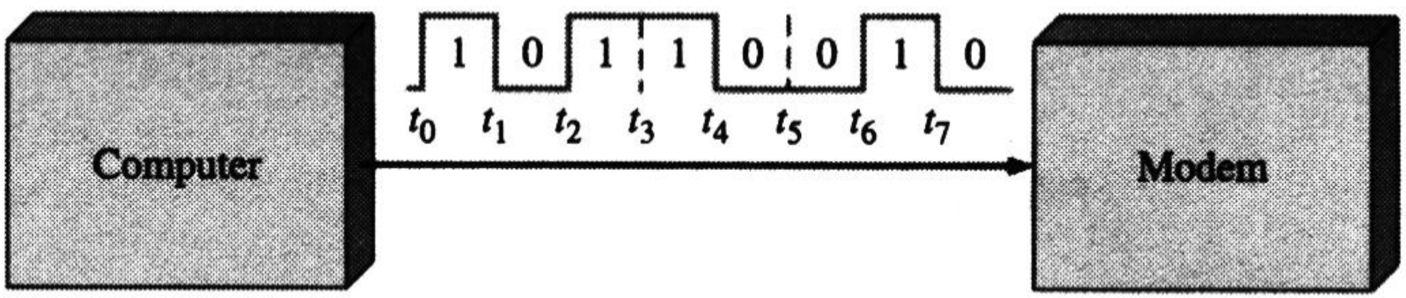
ASCII

TABLE 2–7
American Standard Code for Information Interchange (ASCII).

Control Characters				Graphic Symbols											
Name	**Dec**	**Binary**	**Hex**	**Symbol**	**Dec**	**Binary**	**Hex**	**Symbol**	**Dec**	**Binary**	**Hex**	**Symbol**	**Dec**	**Binary**	**Hex**
NUL	0	0000000	00	space	32	0100000	20	@	64	1000000	40	`	96	1100000	60
SOH	1	0000001	01	!	33	0100001	21	A	65	1000001	41	a	97	1100001	61
STX	2	0000010	02	"	34	0100010	22	B	66	1000010	42	b	98	1100010	62
ETX	3	0000011	03	#	35	0100011	23	C	67	1000011	43	c	99	1100011	63
EOT	4	0000100	04	$	36	0100100	24	D	68	1000100	44	d	100	1100100	64
ENQ	5	0000101	05	%	37	0100101	25	E	69	1000101	45	e	101	1100101	65
ACK	6	0000110	06	&	38	0100110	26	F	70	1000110	46	f	102	1100110	66
BEL	7	0000111	07	´	39	0100111	27	G	71	1000111	47	g	103	1100111	67
BS	8	0001000	08	(	40	0101000	28	H	72	1001000	48	h	104	1101000	68
HT	9	0001001	09	)	41	0101001	29	I	73	1001001	49	i	105	1101001	69
LF	10	0001010	0A	*	42	0101010	2A	J	74	1001010	4A	j	106	1101010	6A
VT	11	0001011	0B	+	43	0101011	2B	K	75	1001011	4B	k	107	1101011	6B
FF	12	0001100	0C	,	44	0101100	2C	L	76	1001100	4C	l	108	1101100	6C
CR	13	0001101	0D	-	45	0101101	2D	M	77	1001101	4D	m	109	1101101	6D
SO	14	0001110	0E	.	46	0101110	2E	N	78	1001110	4E	n	110	1101110	6E
SI	15	0001111	0F	/	47	0101111	2F	O	79	1001111	4F	o	111	1101111	6F
DLE	16	0010000	10	0	48	0110000	30	P	80	1010000	50	p	112	1110000	70
DC1	17	0010001	11	1	49	0110001	31	Q	81	1010001	51	q	113	1110001	71
DC2	18	0010010	12	2	50	0110010	32	R	82	1010010	52	r	114	1110010	72
DC3	19	0010011	13	3	51	0110011	33	S	83	1010011	53	s	115	1110011	73
DC4	20	0010100	14	4	52	0110100	34	T	84	1010100	54	t	116	1110100	74
NAK	21	0010101	15	5	53	0110101	35	U	85	1010101	55	u	117	1110101	75
SYN	22	0010110	16	6	54	0110110	36	V	86	1010110	56	v	118	1110110	76
ETB	23	0010111	17	7	55	0110111	37	W	87	1010111	57	w	119	1110111	77
CAN	24	0011000	18	8	56	0111000	38	X	88	1011000	58	x	120	1111000	78
EM	25	0011001	19	9	57	0111001	39	Y	89	1011001	59	y	121	1111001	79
SUB	26	0011010	1A	:	58	0111010	3A	Z	90	1011010	5A	z	122	1111010	7A
ESC	27	0011011	1B	;	59	0111011	3B	[	91	1011011	5B	{	123	1111011	7B
FS	28	0011100	1C	<	60	0111100	3C	\	92	1011100	5C	|	124	1111100	7C
GS	29	0011101	1D	=	61	0111101	3D	]	93	1011101	5D	}	125	1111101	7D
RS	30	0011110	1E	>	62	0111110	3E	^	94	1011110	5E	~	126	1111110	7E
US	31	0011111	1F	?	63	0111111	3F	—	95	1011111	5F	Del	127	1111111	7F

Serial Data Transfer(串行传输)

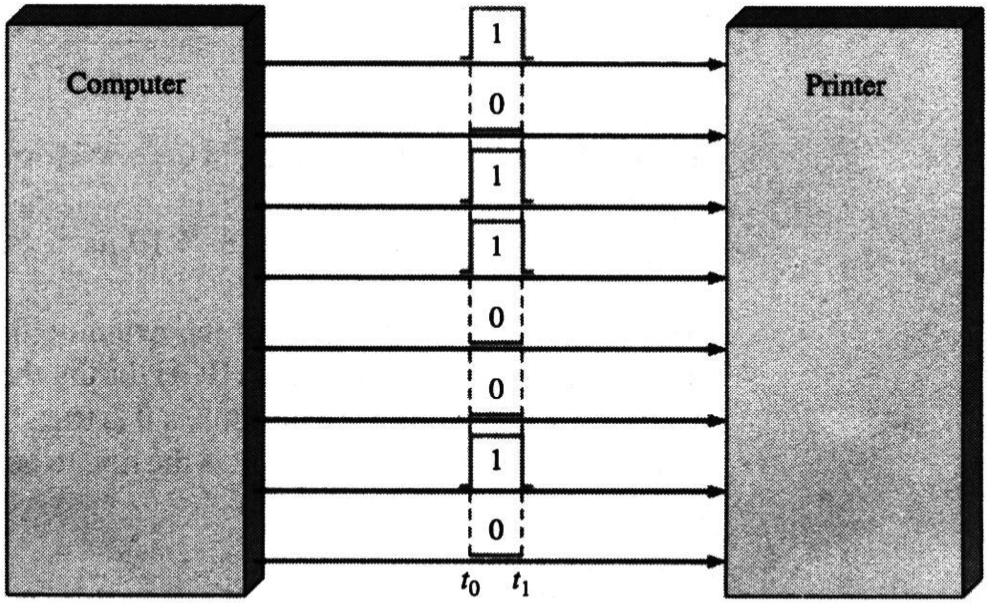
- When bits are transferred in serial form from one point to another, they are sent one bit at a time along a single conductor. To transfer *n* bits in series, it takes *n* time intervals.



(a) Serial transfer of binary data from computer to modem. Interval t_0 to t_1 is first.

Parallel Data Transfer(并行传输)

- When bits are transferred in parallel form, all the bits in a group are sent out on separate lines at the same time. There is one line for each bit. To transfer *n* bits in parallel, it takes one time interval.



(b) Parallel transfer of binary data from computer to printer. t_0 is the beginning time.

FIGURE 1–12
Illustration of serial and parallel transfer of binary data. Only the data lines are shown.

1-3 BASIC LOGIC OPERATIONS

基本逻辑运算

Boolean Algebra（布尔代数）

- In 1850s, the Irish logician and mathematician George Bool developed a mathematical system for formulating logic statements with symbols so that problems can be written and solved in a manner similar to ordinary algebra.

Basic Logic Operations（基本逻辑运算）

- In Boolean algebra, there are three basic logic operations: NOT, AND, and OR.
- Each of the three basic logic operations produces a unique response (output) to a given set of conditions (inputs).
- The standard distinctive shape symbols for the three basic logic operations are shown below.

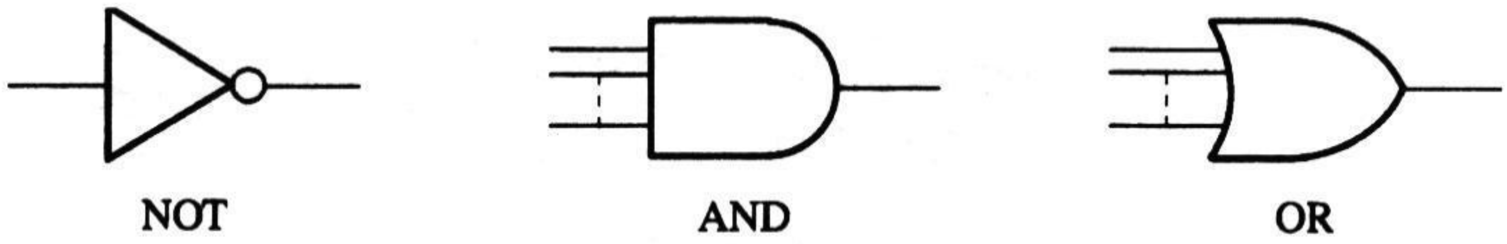
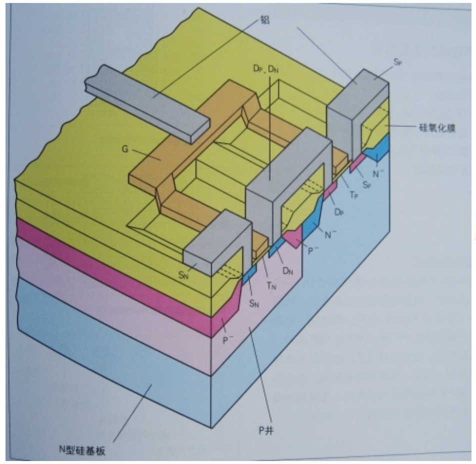
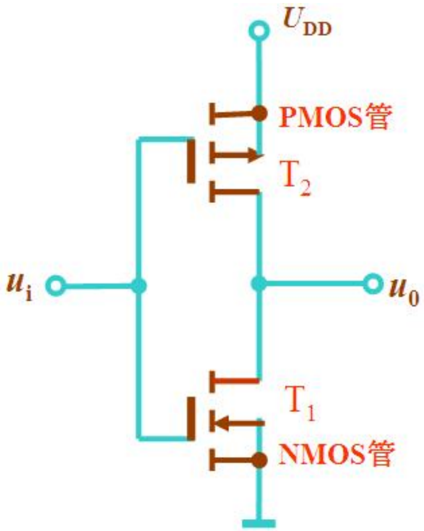


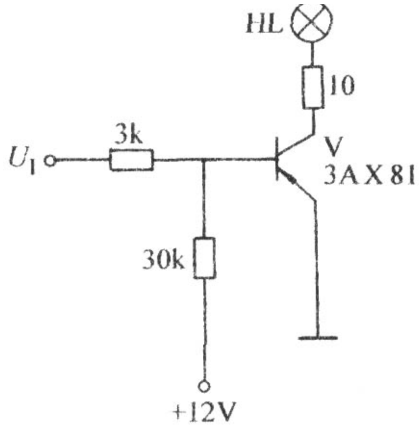
FIGURE 1–15
The basic logic operations and symbols.

The NOT Operation（非运算）

- The NOT operation changes one logic level to the opposite logic level.
- The NOT operation is implemented by a logic circuit known as an inverter (NOT gate).







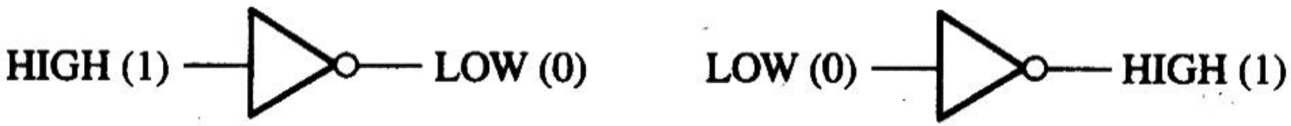


FIGURE 1–16
The NOT operation.

The AND Operation（与运算）

- The AND operation produces a HIGH output only if all the inputs are HIGH. When any or all inputs are LOW, the output is LOW.
- The AND operation is implemented by a logic circuit known as an AND gate.

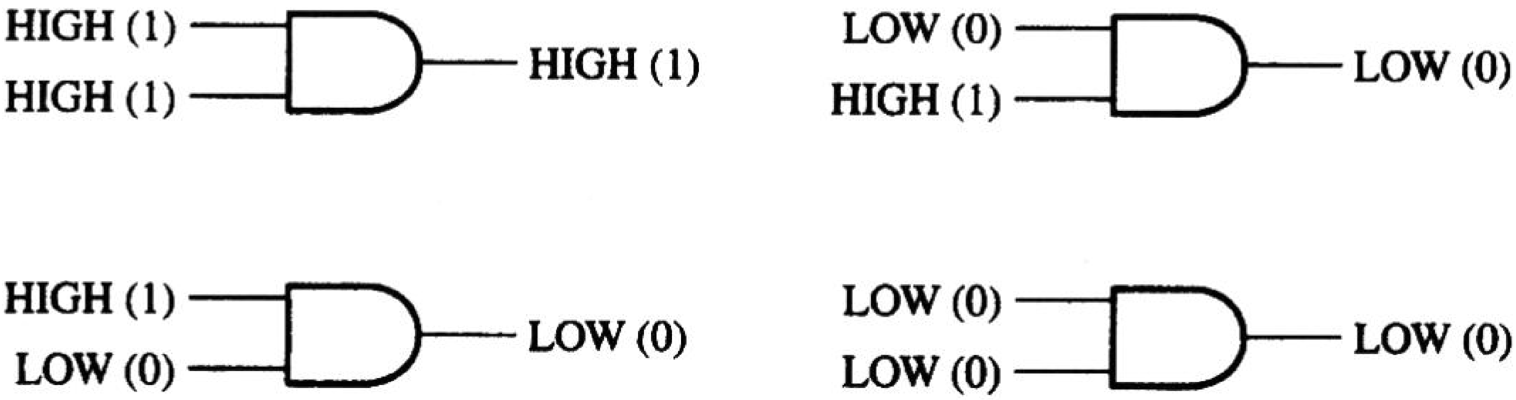
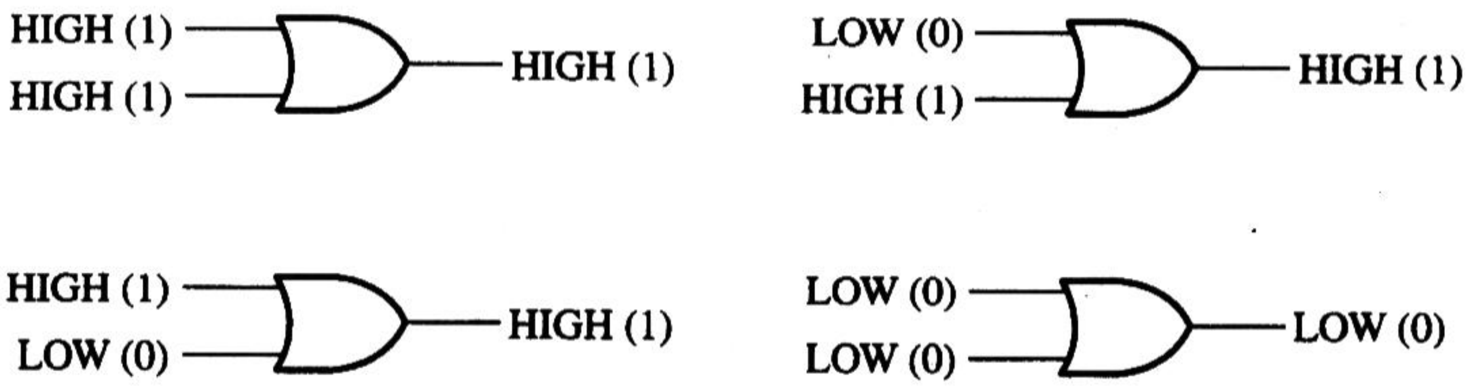


FIGURE 1–17
The AND operation.

The OR Operation（或运算）

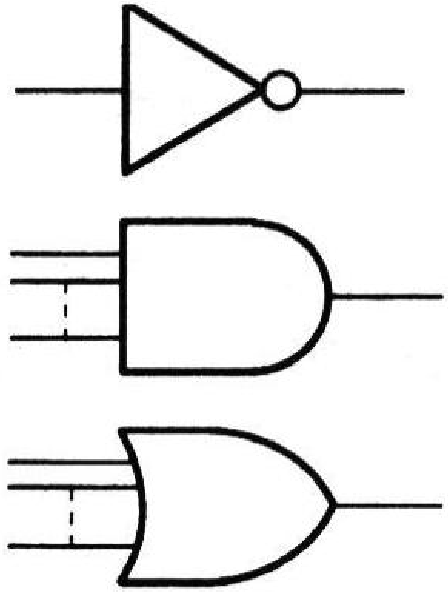
- The OR operation produces a LOW output only if all the inputs are LOW. When any or all inputs are HIGH, the output is HIGH.
- The OR operation is implemented by a logic circuit known as an <u>OR gate</u>.

FIGURE 1–18
The OR operation.



Logic Gates（逻辑门）

- A circuit that performs a specified basic logic operation is called a logic gate. Logic gates form the building blocks for digital systems.
- The true/false statements mentioned earlier are represented by a HIGH (true) and a LOW (false).
- AND and OR gates can have any number of inputs.



1-5 DIGITAL INTEGRATED CIRCUITS

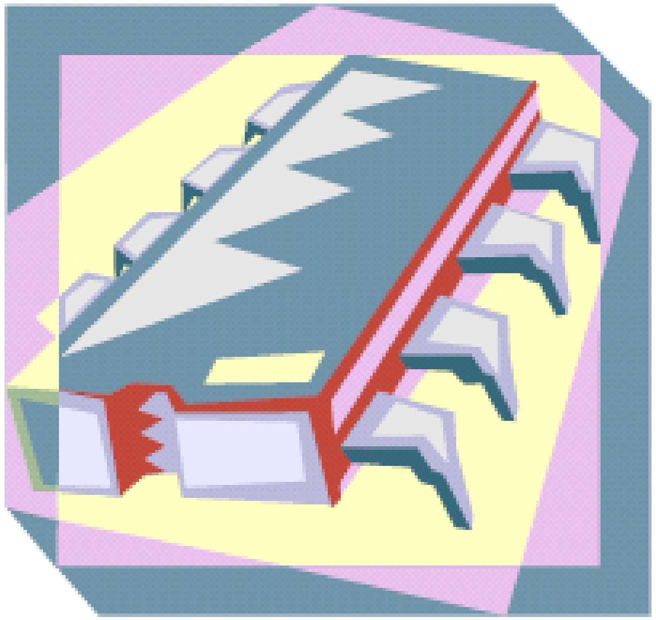
数字集成电路

Introductory Paragraph

- All the logic elements and functions that have been discussed – and many more – are available in integrated circuit (IC) form. Modern digital systems use ICs almost exclusively in their designs because of their small size, high reliability, low cost, and low power consumption. It is important to be able to recognize the IC packages and to know how the pin connections are numbered, as well as to be familiar with the way in which circuit complexities and circuit technologies determine the various IC classifications.
- 集成电路（ integrated circuit ）是目前用的最广泛的一类器件。它具有尺寸小、功耗低、价格便宜、可靠性高等优点。因此学会认识不同芯片的管脚封装类型、使用方法、生产工艺等基本知识十分必要。

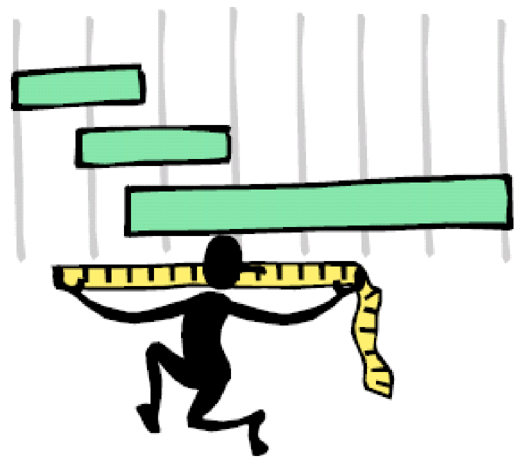
IC

- A monolithic (整体) IC is an electronic circuit that is fabricated on a single chip of semiconductor material (usually silicon) called substrate（基片）. All the components that make up the circuit – transistors, diodes, resistors, and capacitors – are an integral part of that single chip.



Integrated Circuit Complexity Classification

- ICs are classified according to their complexity depending on the number of equivalent gate circuits on a single chip:
  - Small-scale integration (SSI): < 10, gates, flip-flops.
  - Medium-scale integration (MSI): 10 – 100 gates or flip-flops, encoders, decoders, registers, multiplexers (data selectors), adders.
  - Large-scale integration (LSI): 100 – 10000, memories.
  - Very large-scale integration (VLSI): 10^5 – 10^7 , microprocessors, single-chip computers.
  - Ultra large-scale integration (ULSI): 10^7 – 10^9
  - Giga-scale integration (GSI): > 10^9

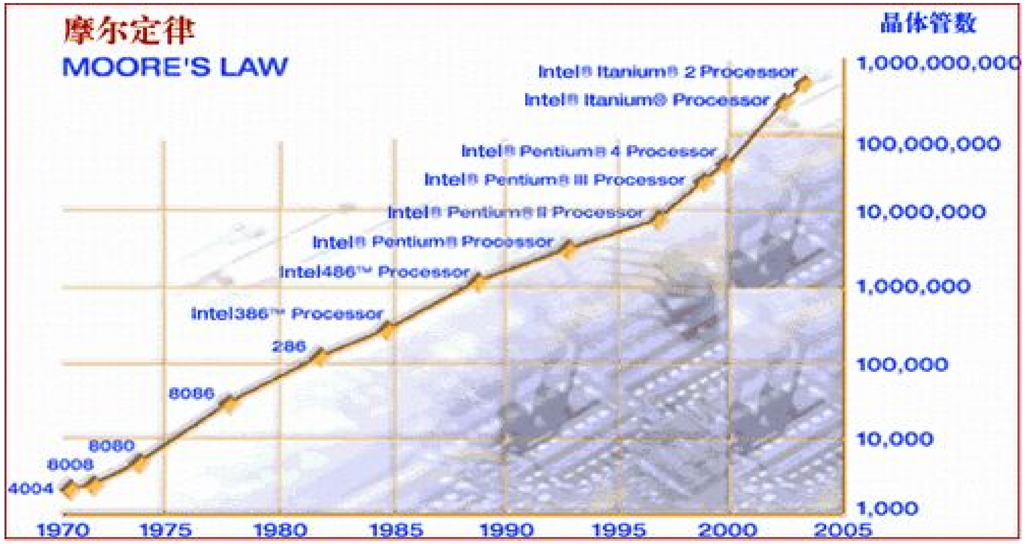


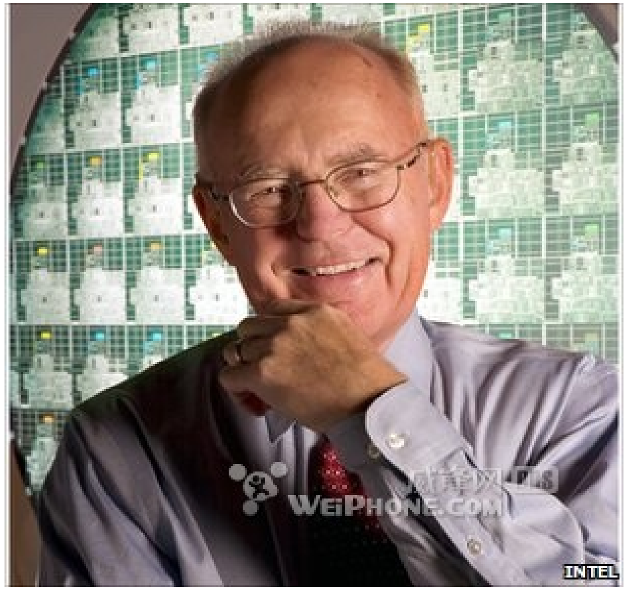
Integrated Circuit Technologies

- The types of transistors with which all ICs are implemented are either BJTs (bipolar junction transistors) or MOSFETs (metal-oxide semiconductor field-effect transistors).
- Two types of digital circuit technology that use BJTs are TTL (transistor-transistor logic) and ECL (emitter-coupled logic).
- The major circuit technologies that use MOSFETs are CMOS (complementary MOS) and NMOS (n-channel MOS).
- SSI and MSI circuits are generally available in both TTL and CMOS.
- LSI, VLSI, ULSI, and GSI are generally implemented with CMOS or NMOS because it requires less area on a chip and consumes less power.

摩尔定律

- 当价格不变时，集成电路上可容纳的元器件的数目，约每隔18-24个月便会增加一倍，性能也将提升一倍。





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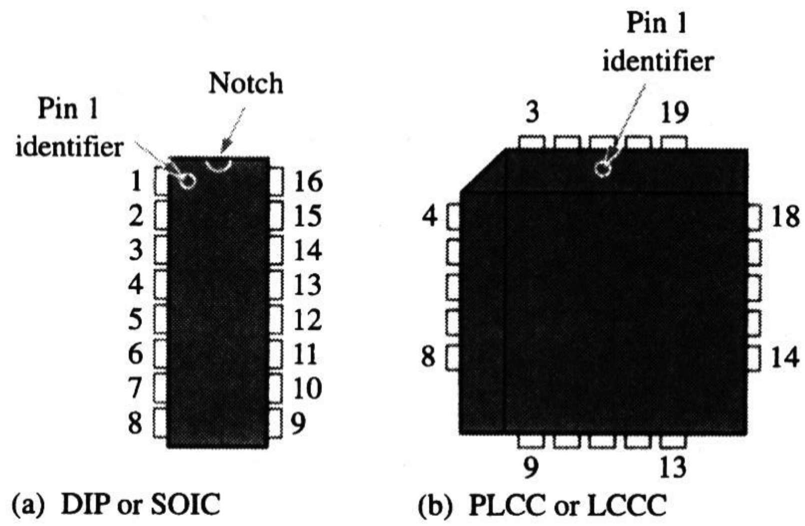
Pin Numbering（引脚编号）

- All IC packages have a standard format for numbering the pins. Looking at the top of the package, pin 1 is indicated by an identifier that can be either a small dot, a notch, or a beveled edge. Starting with pin 1, the pin numbers increase as you go counterclockwise.

FIGURE 1–30
Pin numbering for two standard types of IC packages. Top views are shown.



Quiz

1. Compared to analog systems, digital systems

   a. are less prone to noise

   b. can represent an infinite number of values

   c. can handle much higher power

   d. all of the above

Quiz

2. The number of values that can be assigned to a bit are

   a. one

   b. two

   c. three

   d. ten

Quiz

3. The time measurement between the 50% point on the leading edge of a pulse to the 50% point on the trailing edge of the pulse is called the

   a. rise time

   b. fall time

   c. period

   d. pulse width

Quiz

4. The time measurement between the 90% point on the trailing edge of a pulse to the 10% point on the trailing edge of the pulse is called the

   a. rise time

   b. fall time

   c. period

   d. pulse width

Quiz

5. The reciprocal of the frequency of a clock signal is the

   a. rise time

   b. fall time

   c. period

   d. pulse width

Quiz

6. If the period of a clock signal is 500 ps, the frequency is

   a. 20 MHz

   b. 200 MHz

   c. 2 GHz

   d. 20 GHz

Quiz

7. AND, OR, and NOT gates can be used to form

   a. storage devices

   b. comparators

   c. data selectors

   d. all of the above

Quiz

9. A device that is used to switch one of several input lines to a single output line is called a

   a. comparator

   b. decoder

   c. counter

   d. multiplexer

Quiz

10. For most digital work, an oscilloscope should be coupled to the signal using

a. ac coupling

b. dc coupling

c. GND coupling

d. none of the above

Homework

- Problems: 1, 2, 6(b), 7, 8, 12, 13, 17